

Ricky Ding

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Entrepreneur, Full-Stack Developer, Physics Researcher

Visit <https://rickyding1997.github.io> for detailed descriptions, screenshots, certificates and latest CVs

Education

National University of Singapore

Bachelor of Science (Honours, Highest Distinction) in Physics

Jun 2020

- **Thesis:** Density Potential Functional Theory in position and momentum space and its implementation using PyTorch
- Developed original orbital-free DPFT code with multi-GPU acceleration using PyTorch
- Implemented 2D/3D magnetic dipole-dipole interactions; compared results with VASP and custom Kohn-Sham DFT
- Science & Technology Undergraduate Scholarship recipient
- [Certificates and thesis available online](#)

University of Göttingen, Germany

Non-Degree Student (Study Abroad)

Summer 2019

- Relevant coursework: Biomolecular Physics and Simulations (1.3/1.0), Advanced Topics in Biophysics (1.0/1.0)
- German grading scale: 1.0 (best) – 4.0 (pass)

Technical Skills

- **Programming:** Python, C++, JavaScript, SQL, Verilog
- **Scientific/AI:** PyTorch, Numba, NumPy, Scikit-learn, Pandas
- **Trading Systems:** Backtesting engines, optimization algorithms, market data pipelines
- **Frontend:** Vue, React, React Native
- **Backend:** Bun, Node.js, Elysia, FastAPI, PostgreSQL, Redis
- **DevOps:** Docker, Caddy, Linux

Quantitative Trading Experience

Serena Trading Research (Personal Project)

Founder & Developer

2025 – Present

- Developed [crypto-monitor](#): Real-time monitoring of entire crypto market with concurrent data streaming
- Built [crypto-data-downloader](#): Downloads 1 month of market data (597 symbols, 5min intervals) in 3 minutes
- Created [fast-trading-simulator](#): Numba-accelerated minimalist simulator with multi-symbol, multi-position support
- Implemented [trading-models](#): Neural network models (MLP, CNN, Transformer) for trading signals
- Developed [minimalist-RL](#): Minimalist reinforcement learning framework for strategy optimization
- Designed [grad-free-optim](#): Gradient-free optimization using genetic algorithms for parameter tuning

Yinhan Tradebot (Archived Project)

Developer & Trader

Apr 2024 – Jul 2025

- Developed and deployed personal trading bot achieving +300% profit with -28% max drawdown
- Converted to multi-user web app supporting real-time trading and portfolio management
- Ran algorithm MSB1 for 6 months achieving +600% profit before -30% drawdown
- Key learning: Risk management and drawdown minimization are more critical than profit maximization
- [Detailed performance metrics and analysis available](#)

Tesla Coil Physics Simulator

Researcher & Developer

2023 – Present

- Developed [tesla-coil-simulator](#): Physics simulator for DRSSTC (coupled series RLC resonators)
- Implemented mathematical modeling of complex coupled oscillator systems
- Applied numerical methods for solving differential equations in high-frequency systems
- [Research note and results available](#)

Other Relevant Experience

[FRONTLINE INDUSTRIAL SOFTWARE PTE. LTD](#)

CTO & Co-Founder

Nov 2021 – Present

- Received USD \$700K seed investment from Cocoon Capital and Plug and Play (2023)
- Developed novel differentiable optimization algorithm for project scheduling
- Transformed non-differentiable scheduling problems into differentiable computational graphs
- Applied gradient descent and backpropagation to large-scale optimization problems
- Built maintenance scheduling system using Mixed Integer Linear Programming
- Implemented overrun forecasting using NLP vectorization and HDBSCAN clustering

[Centre for Quantum Technologies, NUS](#)

Researcher

Jul 2020 – May 2021

- Quantum Optimal Control using Deep Reinforcement Learning
- Designed DRL environment using OpenAI Gym for quantum state control
- Implemented SAC algorithm on qubit and 3-level Transmon systems
- Developed systematic calibration procedure for IQ mixer minimization
- Automated tuning process for quantum measurement systems

Entrepreneurial Projects

- [EntreGames](#): AI-powered entrepreneurship simulation platform
- [Similar Minds](#): Platform for finding people with similar ideas using vector similarity search
- Both projects feature full-stack development with AI integration and real-time systems

Updated: December 16, 2025 | Version: Quantitative Trading